

9

Fundamentals of Controller Performance Assessment

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Until now we have been focused on designing controls for SISO and MIMO systems. Traditionally, the process engineer's job was to tune controllers. However, nowadays, with sophisticated distributed control systems (DCS) in place, tuning is usually performed by instrumentation engineers who deploy DCS in plants. Once DCS is deployed, the job of the process engineer is to ensure that the system performs adequately to achieve process objectives. However, like all systems, performance of the control system also degrades with time. This deterioration can be attributed to many reasons and ultimately manifests as poorly performing control loops. In a typical process plant, there are hundreds of control loops. It is the control engineer's job to monitor these loops.

Few engineers monitor all the loops, making manual monitoring error-prone and difficult. Hence, automated approaches for monitoring control loops are essential and likely to have a significant impact. In this chapter, we will study this area and describe important ideas in the field of controller performance monitoring and assessment.

In the previous chapters, design of controllers given a transfer function of the process was described. This is relevant and critical at the time of installation of controllers. During initial installation, experimentation with the plant is allowed so that process models are built for each loop and controllers tuned based on the process model. However, over a period of time, the process may drift away from its observed initial behaviour due to various reasons. For example, physical performance of the equipment can degrade through wear and tear, catalysts can get deactivated, there could be scaling in heat exchangers, and so on. As a result, the original models that were built during controller commissioning might not be valid after a period of time. Performing perturbation experiments post commissioning of controllers is generally difficult as there are other factors such as production loss that need to be considered. It is also difficult to maintain process models of every control loop in a plant as there may be hundreds to thousands of control loops.

This situation, where one needs to assess the performance of the control loops without reliable process models has spawned a new research area, popularly termed as controller performance assessment and diagnosis (CPAD).

The aim of this chapter is to describe this area of control in a structured manner. The problem is the following. Consider the closed-loop described in Figure 9.1. For each of these loops, the following data is generally available:

1. Setpoint profile (y_{sp}) SP (setpoint) or SV (set value) in DCS notation
2. Process response (y) or PV (process value) in DCS notation
3. Controller output (u) or MV (manipulated value) or OP(Output) in DCS notation
4. Controller parameters K_c, τ_I, τ_D if the controller is a PID controller

With this information, one would like to answer:

1. Given the current state of the process, is the controller functioning optimally?
2. If it is not performing optimally, what is the reason for sub-optimal performance?

In general, y (PV) should be as close to y_{sp} (SP) as possible. However, the output might not follow the setpoint in many cases. As shown in Figure 9.2, the observed deviations could be categorized as: (i) drifts, (ii) poor dynamics, (iii) high variability, and (iv) oscillations. The aim of monitoring techniques

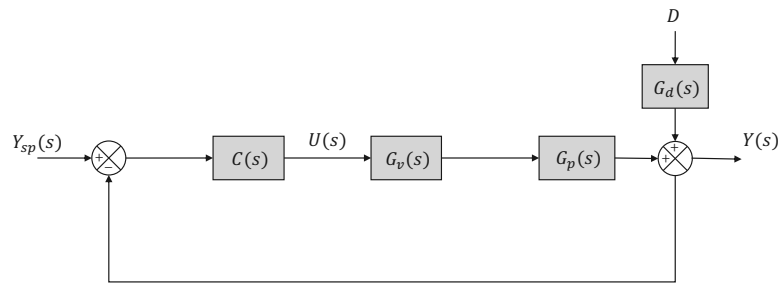


FIGURE 9.1: Typical closed-loop system

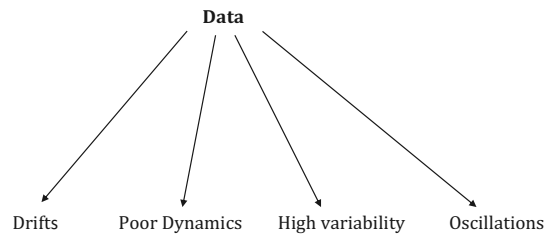


FIGURE 9.2: Features in data from poorly performing control loops

is the automatic identification of these features in control loop data so that problems in controllers can be addressed expeditiously.

Consider the output of four linear systems with PID controllers. Plots in Figure 9.3 indicate data with the four features described above: drift, poor dynamics, high variability, and oscillations around setpoint. From these plots, it seems as if it is simple to visually identify problems in control loops. A drift from setpoint can be seen by plotting the setpoint and the process variable in the same graph. Similarly oscillations can also be visually detected. However, this is generally not possible in a real plant scenario. First, in a plant with hundreds of loops and a few control engineers, manual analysis of all control loop data is virtually impossible. Further, there are limitations on the display length and number of variables that can be rendered in a DCS screen.

As a result, there is a need for automated techniques to identify such anomalous behaviour. Interestingly, what might be visually simple to identify is, in general, difficult to automate. Once the problem is identified, then the root cause for these problems should be diagnosed for further corrective action. For example, if the output is oscillatory around the setpoint, how does one

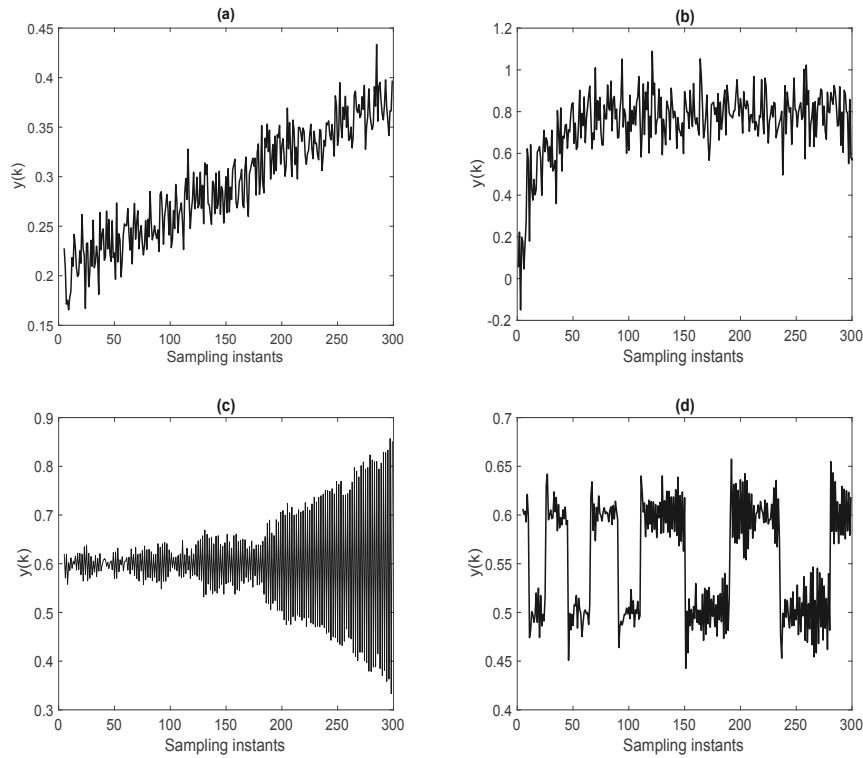


FIGURE 9.3: Plots from linear closed-loop systems (time v/s amplitude) exhibiting (a) drifts, (b) poor dynamics, (c) high variability, and (d) oscillations.

rectify the problem? It turns out that the approach to reduce oscillations would depend on the root cause for oscillations. As a result, the first step in mitigation is the diagnosis of the cause for poor performance. Similar remarks can be made about variability and drifts in the process. Finding the root cause for anomalous behaviour is called control loop performance diagnosis.

9.1 Performance Assessment of Control Loops

In this chapter, techniques that automatically detect the four features in control loop data are described. While there are several techniques that are available, simplicity, importance, and consistency with the material presented in this book are important considerations for identifying material to be presented

in this chapter. Further, a mix of model-based and data-based methods have been chosen for presentation.

9.2 Identifying Drifts

Data that consists of an ordered set of values at different sampling instants is called a time series. Time series data for variables related to control loops such as PV and SP are usually available. In simple terms, drift between two variables is defined as a consistent and increasing separation between these variables over time. Drift in an output variable is measured as the deviation of this output from its setpoint. Drift in an output, if it exists, should be visible by plotting the time series data for error, $e = y_{sp} - y = SP - PV$.

Consider Figure 9.4. In Figure 9.4a, SP , PV and e are plotted in the same graph. This is the case of no drift between SP and PV . Due to measurement errors (which are unavoidable), e values are never equal to zero. As a result, thresholding is needed to avoid detecting drifts when there are none. In Figure 9.4b, the same time period with a very slight drift is depicted. In this case, it is not easy, even visually, to identify drift in the process variable. Drift becomes visible only if the time period is expanded to include a large number of data points. In Figure 9.4c, we show the same time period with a strong drift. In this case it is quite easy to see the drift in the error plot. In summary, identifying drifts visually might not be simple in all cases. Further, it is difficult to manually monitor hundreds of variables to identify drifts. Algorithms that automatically identify drift in data will be quite useful. One of the simplest techniques for this is called the CUSUM chart, where CUSUM stands for cumulative sum.

Consider the time series $e(k)$ over a period of time. Instead of plotting the error $e(k)$ over time, let us plot the cumulative sum of $e(k)$ over the time period. At time $t = k$, compute:

$$\text{CUSUM}(k) = \sum_{i=0}^k e(i) \quad (9.1)$$

When this is plotted, the result is seen in Figure 9.5 for the three cases. In the first case, CUSUM hovers around zero, whereas in the other two cases the value deviates from zero, with case three rate of change being much higher than case two. If one were to use the dotted lines shown in Figure 9.5 as thresholds for detection, then the drift could be easily detected in both cases, while minimizing false detection due to measurement noise.

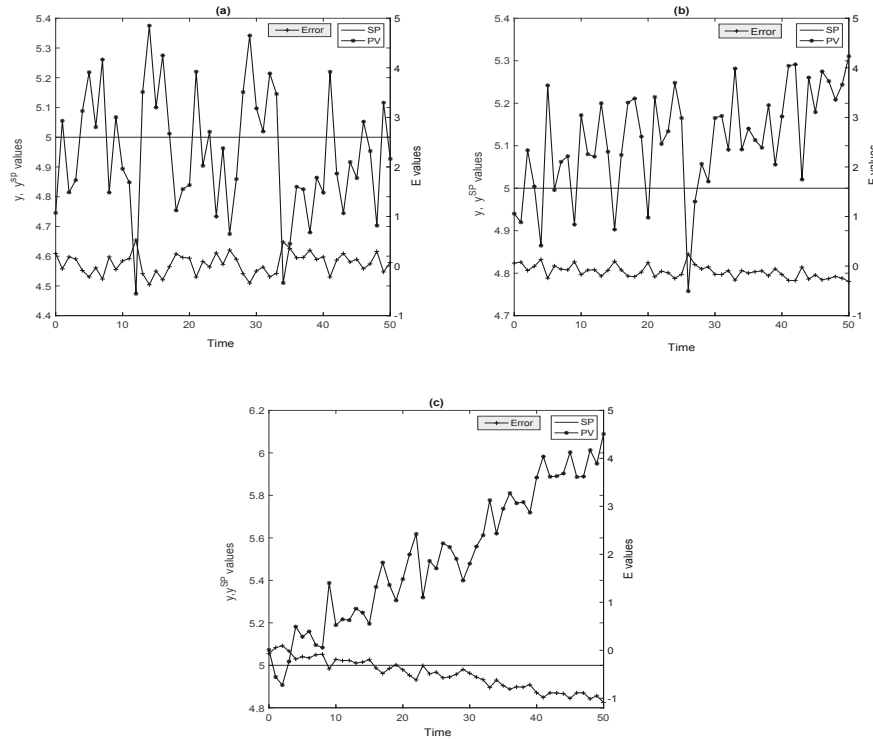


FIGURE 9.4: (a) Data without drift, (b) data with little drift, and (c) data with significant drift

9.3 Identifying Poor Dynamics

As described until now in the book, the goal of a control system is to follow setpoint changes (servo) and reject disturbances (regulatory control). If one were to think of tuning of controllers as designing for the two control objectives, then controller performance assessment (CPA) can be thought of as verification of performance, once controllers are implemented. In this view, performances of controllers have to be assessed for: (i) **step in setpoint**, (ii) **step disturbances**, and (iii) **non-step type disturbances**. One technique for each of the cases (i) and (ii) and two techniques for case (iii) are described in this chapter. While these techniques can be modified and applied for other cases, to keep the presentation simple, we focus on a technique only for a particular case. Oscillations in control loops is addressed as a special case because oscillations can be a result of poor controller tuning and/or due to nonlinearities in control valves. We know that if the controller parameters are such that the

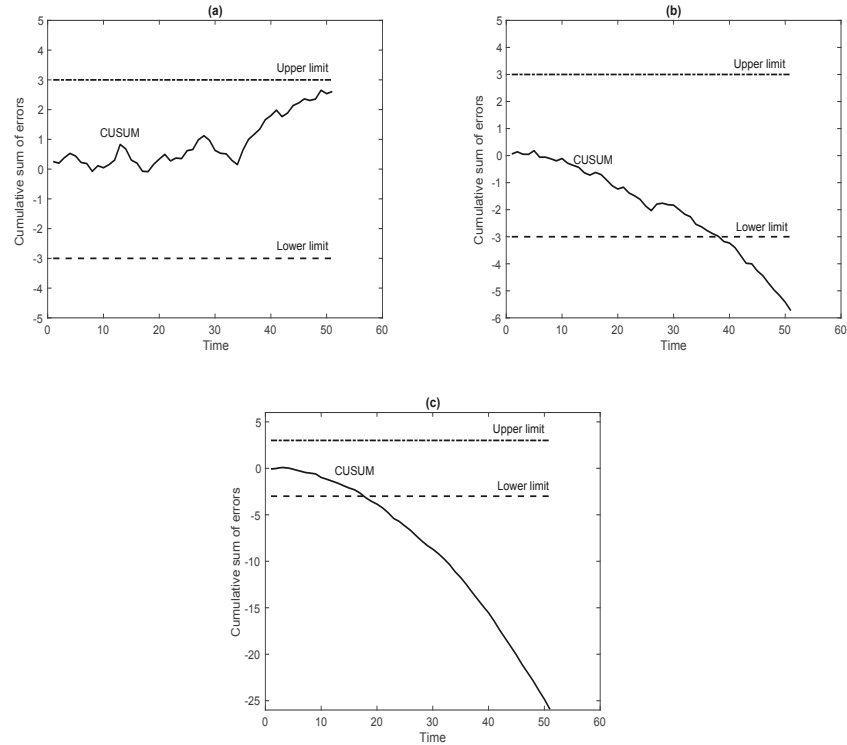


FIGURE 9.5: (a) CUSUM plot for data given in Figure 9.4a, (b) CUSUM plot for data given in Figure 9.4b, and (c) CUSUM plot for data given in Figure 9.4c

closed-loop poles are close to the imaginary axis, significant oscillations are likely. This could happen because the process has changed over time and the tuning with the original back-off is not optimal anymore for the new process. The reasons for valve nonlinearities inducing oscillations will be described in detail in the next chapter. Notwithstanding all this, it is important to monitor control loops for oscillations and we describe one algorithm for oscillation detection in this chapter.

Monitoring algorithms are broadly classified into two categories: model-based and data-based. The strategy that the monitoring algorithms employ, in almost all of these cases, can be summarized as follows. In the category of model-based techniques, some reference model or parameters related to the control loop is provided to the algorithm. These algorithms use online control loop data (PV , SP and OP) to identify certain quantities that can be computed from the data. The computed quantity is compared with the theoretically derived value of the quantity based on the model and the expected

performance of the closed-loop for the given model. These computed and theoretical values are used to derive a benchmark, which allows the detection of poor performance. When the technique is purely data-based, model-free benchmarks based on a conceptual understanding of the expected behaviour of systems that are optimal is used. There could be quite a bit of subjectivity in these cases.



9.3.1 Model-based Approach - DS/IMC Tuning Rule Based Indices

An algorithm for the first case, *i.e.*, performance of the closed-loop when there are step changes in the setpoint is described. This is a model-based technique and the expected performance of the closed-loop is based on the D-S approach, where a G^{des} is used. Complete details about this technique can be found in Veronesi and Visioli [97]. We present a slightly simplified version of this algorithm.

- Assume that a model of the process that is second-order of the form $G_M(s) = \frac{K_p e^{-\tau_d s}}{(\tau_1 s + 1)(\tau_2 s + 1)}$ or higher order is available.
- If the model is of a higher order then convert the model to the second-order form shown above using the following heuristic:

$$\tau_1 = \tau_{10}, \quad \tau_2 = \tau_{20} + \frac{\tau_{30}}{2}, \quad \tau_d = \tau_{d0} + \frac{\tau_{30}}{2} + \sum_{i \geq 4} \tau_{i0} \quad (9.2)$$

Here τ_{10} , τ_{20} and τ_{i0} are the time constants of the process arranged in descending order, with τ_{10} being the largest time constant. τ_{d0} is the delay in the higher order model and τ_d is the delay in the approximated second-order plus time-delay (SOPTD) model. Beyond the first two dominant time constants, the lag due to other time constants are lumped into the delay (τ_d) element in the SOPTD model. For lumping the lag due to time constants, use the approximation rule: $e^{-\tau_d s} \approx \frac{1}{\tau_d s + 1}$, the small time constants along with the process delay in the original model is added to form the overall delay τ_d of the SOPTD model.

- Define a variable T_0 which is the sum of all the time constants and dead time of the process which is given by:

$$T_0 = \tau_1 + \tau_2 + \tau_d = \left(\tau_{d0} + \sum_i \tau_{i0} \right) \quad (9.3)$$

- Once the SOPTD system approximation is complete, perform a DS-based controller transfer function identification with the objective that closed-loop transfer function should behave like a first-order system with time-delay and time constant τ_c (as controller action can be realized in the process output only after the delay period). In the Skogestad

Internal Model Control (SIMC) tuning rule, the time constant of the closed-loop system is set equal to the time-delay of the process. Using the direct synthesis approach (Equation (6.16)) with $G^{des} = \frac{e^{-\tau_d s}}{\tau_c s + 1}$ and $G_M = K_p \frac{e^{-\tau_d s}}{(\tau_1 s + 1)(\tau_2 s + 1)}$ we have:

$$C(s) = \frac{(\tau_1 s + 1)(\tau_2 s + 1)}{K_p} \frac{1}{(\tau_c s + 1 - e^{-\tau_d s})} \quad (9.4)$$

Here τ_c is the closed-loop time constant. On applying the approximation for $e^{-\tau_d s} = 1 - \tau_d s$ and equating $\tau_c = \tau_d$, we have:

$$C(s) = \frac{(\tau_1 s + 1)(\tau_2 s + 1)}{K_p} \frac{1}{(2\tau_d) s} \quad (9.5)$$

- The above equation is a PID controller with settings given by (where the controller is of the series form):

$$C(s) = K_c \left(\frac{\tau_I s + 1}{\tau_I s} \right) (\tau_D s + 1), \quad K_c = \frac{\tau_1}{2K_p \tau_d}, \quad \tau_I = \tau_1, \quad \tau_D = \tau_2 \quad (9.6)$$

With these settings, the variable T_0 can be written as:

$$T_0 = \tau_I + \tau_D + \frac{\tau_I}{K_c K_p} \quad (9.7)$$

- The PID controller with above settings in closed-loop will result in the closed-loop transfer function:

$$G^{cl}(s) = \frac{G_M(s)C(s)}{1 + G_M(s)C(s)} = \frac{e^{-\tau_d s}}{\tau_d s + 1} \quad (9.8)$$

- The integral absolute error (IAE) defined as:

$$IAE = \int_0^\infty |e(t)| dt \quad (9.9)$$

can be shown to be $2\tau_d$ for a unit step in setpoint. The value for IAE can be obtained by considering unit step response of the closed-loop system $G^{cl}(s)$. Then:

$$Y(s) = \frac{1}{s(1 + \tau_d s)} e^{-s\tau_d} \quad (9.10)$$

Using partial fraction expansion and taking inverse LT, $y(t)$ is:

$$y(t) = \alpha_{\tau_d}(t) - e^{-\frac{t-\tau_d}{\tau_d}} \alpha_{\tau_d}(t)$$

where $\alpha_{\tau_d}(t)$ is a step starting at time $t = \tau_d$. The error signal $e(t)$ for a unit step is given by:

$$e(t) = y_{sp} - y(t) = 1 - y(t) \quad (9.11)$$

IAE can be computed as:

$$\begin{aligned} IAE &= \left| \int_0^\infty \left(1 - \alpha_{\tau_d}(t) + e^{-\frac{t-\tau_d}{\tau_d}} \alpha_{\tau_d}(t) \right) dt \right| \\ &= \int_0^{\tau_d} dt + \int_{\tau_d}^\infty e^{-\frac{(t-\tau_d)}{\tau_d}} dt \\ &= \tau_d + \int_{\tau_d}^\infty e e^{-\frac{t}{\tau_d}} dt \\ &= \tau_d + e \int_{\tau_d}^\infty e^{-\frac{t}{\tau_d}} dt \\ &= \tau_d - e\tau_d \left[e^{-\frac{t}{\tau_d}} \right]_{\tau_d}^\infty \\ &= \tau_d - e\tau_d(0 - e^{-1}) \\ &= 2\tau_d \end{aligned}$$

- If the closed-loop system has an SIMC controller, then IAE of the closed-loop process for a unit step change in setpoint is $2\tau_d$. This information is used to develop SIMC based controller performance index, which is described next.

9.3.1.1 Algorithm for Development of SIMC Based Performance Indices

SIMC based performance assessment indices are developed so that they can be calculated with only the setpoint response data along with the knowledge of the controller parameters. The detailed algorithm is presented below:

- Assume that closed-loop data for a unit step change in the setpoint is available
- The steady-state gain of the process K_p can be obtained by computing the final value of the controller signal $u(t)$. Obtain the final value of controller signal $u(t)$ using $\lim_{t \rightarrow \infty} u(t) = \frac{K_c}{\tau_I} \int_0^\infty e(t) dt = \frac{1}{2\tau_d K_p} \int_0^\infty e(t) dt = \frac{1}{K_p}$ for a unit step in the setpoint using final value theorem.

From this relation we have the steady-state gain of the process (based on closed-loop step response) as:

$$K_p = \frac{\tau_I}{K_c \int_0^\infty e(t) dt} \quad (9.12)$$

Here K_c and τ_I are the gain and integral time constant respectively of the PI controller.

- The variable T_0 (net effect of time constants and delay in the process) is calculated from the process data by considering $x(t) = K_p u(t) - y(t)$ where $u(t)$ is the controller output, $y(t)$ is the process output and K_p is the process gain calculated from the previous step. Now computing $\int_0^\infty x(t) dt$ provides the value of T_0 .
- $\int_0^\infty x(t) dt$ can be derived by applying FVT (considering the setpoint as unit step):

$$\begin{aligned} \lim_{s \rightarrow 0} s \frac{1}{s} X(s) &= \lim_{s \rightarrow 0} s \frac{1}{s} \frac{C(s) [K_p - G_M(s)]}{1 + G_M(s)C(s)} Y_{sp}(s) \\ &= T_0 \end{aligned} \quad (9.13)$$

- The values computed from the data can be compared with the theoretically derived values given by the following indices, where the denominators are computed using data and current controller parameters:

$$\text{Sigma index (SI)} = \frac{\tau_I + \tau_D + \frac{\tau_I}{2K_p K_c}}{T_0} \quad (9.14)$$

$$\text{Closed-loop index (CI)} = \frac{2\tau_d}{\int_0^\infty |e(t)| dt} \quad (9.15)$$

- If the control loop is tuned according to SIMC, the value of SI should be one. If SI value is less than one, then the controller parameters are tuned aggressively. The closed-loop behaviour can be improved by reducing the controller gain and increasing the integral time constant. However, a value greater than one suggests a sluggish controller and can be improved by increasing the controller gain and decreasing the integral time constant.
- If SIMC based tuning is implemented, CI value should be one. However, PID controllers are tuned conservatively in industries (to avoid aggressive action resulting in damage to control valves) and therefore, a CI value greater than 0.6 is desirable. If CI value is below 0.6, the controller performance is poor and it is good to re-tune the controller.

From Figure 9.6 (summary of the algorithm), it can be clearly seen that the performance indices are computed using only the controller output, process output, controller parameters, and a nominal model. While the algorithm is based on an SOPTD model because of its wide use, the algorithm can be used, with minor modifications, for FOPTD models. The series form assumed for process models can also be relaxed through the use of appropriate translation formulae [98].

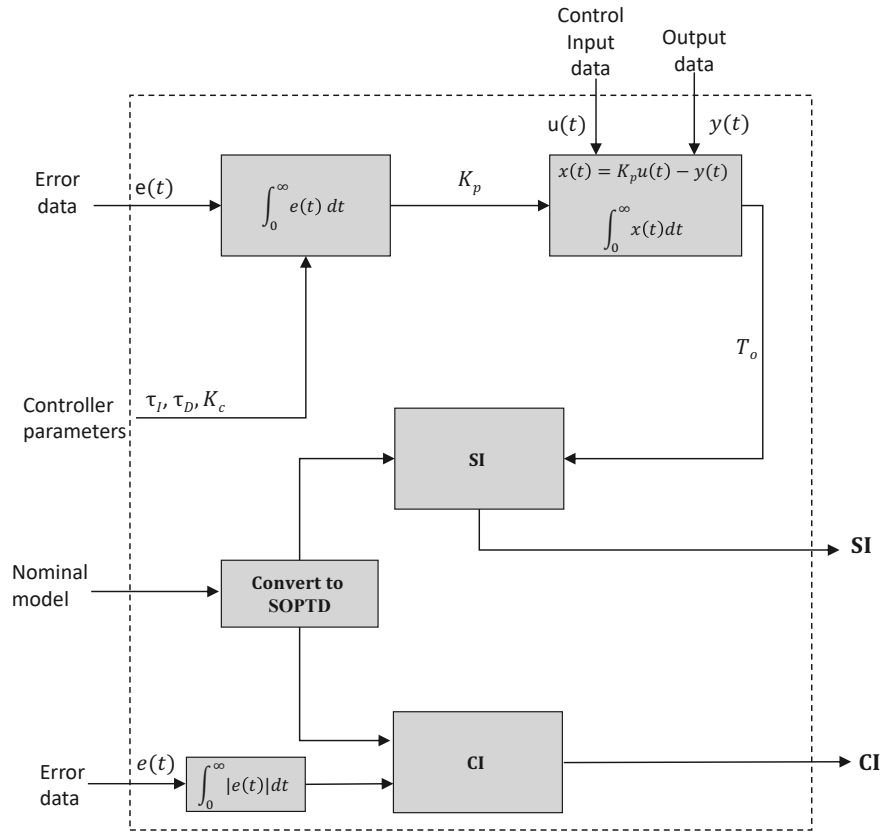


FIGURE 9.6: Flowchart of the algorithm to monitor controller performance in response to changes in setpoints

9.3.2 Idle Index Technique for Identification of Sluggish Control Loops

We turn our attention now to performance assessment in the case of step type load disturbance. Idle index [31] has been proposed as a metric for quantifying less than optimal rejection of step disturbances by identifying the presence or absence of sluggish control response. This is a model-free approach and hence a performance measure is developed based on an intuitive understanding of closed-loop behaviour when step disturbances occur. As the name suggests, this index quantifies the fraction of time during which the controller is deemed to have been idle (or lagging) when it should have been counteracting the disturbance. This idleness is quantified by assessing simultaneous variations in controller and process outputs when the process is subjected to load changes. Ideally, when the process output is increasing (decreasing), one would expect

the controller output to counteract that by decreasing (increasing). It can be surmised from this that the rate of change of process and controller outputs should be monitored to identify idleness.

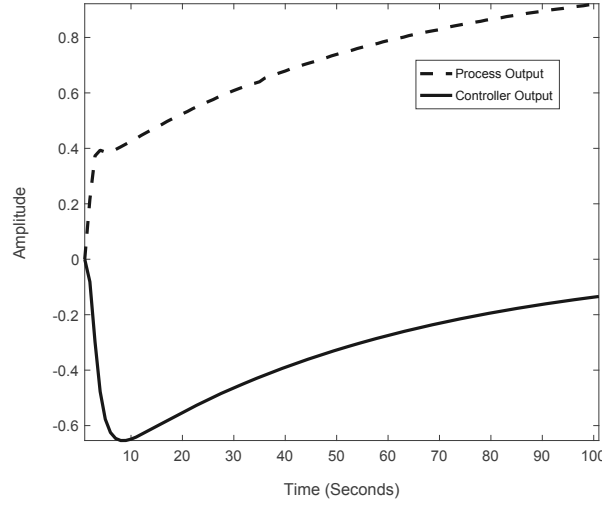


FIGURE 9.7: Response of a closed-loop system with PID controller setting $K_c = 0.5$, $\tau_I = 20$ and $\tau_D = 2$ for a load change of one

Consider Figure 9.7, where the controller and output responses are plotted for the process $G(s) = \frac{1}{(s+1)^3}$ in a closed-loop with PID controller setting $K_c = 0.5$, $\tau_I = 20$ and $\tau_D = 2$ for an unit load change. Both the controller and process outputs (u & y) have an initial change (as seen from Figure 9.7) with $\Delta u \Delta y < 0$, and there is a long period during which both $\Delta u \Delta y > 0$. For the same system with $K_c = 2.5$, $\tau_I = 1$ and $\tau_D = 1$, though there is an initial negative period with $\Delta u \Delta y < 0$, there is no longer a positive relation between Δu and Δy as shown in Figure 9.8.

The characteristic feature of a sluggish control loop is the long period of time during which $\Delta u \Delta y > 0$ (both of them follow the same trend). This sluggishness is quantified through the calculation of two quantities, denoted t_{pos} and t_{neg} . The computation is as follows. Identify a time period and collect data for two time series $y(k)$ and $u(k)$. Let T_s be the sampling period. If we recall, $t = k$ is short hand notation for $t = kT_s$. Set $t_{pos} = t_{neg} = 0$. With this initialization, scan every sampling instant in the time series data sequentially and compute:

$$t_{pos} = \begin{cases} t_{pos} + T_s & \text{if } \Delta u \Delta y > 0 \\ t_{pos} & \text{if } \Delta u \Delta y \leq 0 \end{cases} \quad (9.16)$$

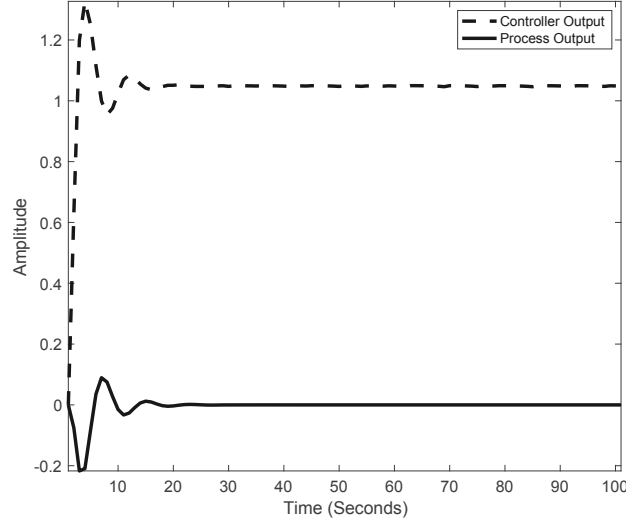


FIGURE 9.8: Response of a closed-loop FOPTD system with PI controller setting $K_c = 2.5$, $\tau_I = 1$ and $\tau_D = 1$ for a load change of one

$$t_{neg} = \begin{cases} t_{neg} + T_s & \text{if } \Delta u \Delta y < 0 \\ t_{neg} & \text{if } \Delta u \Delta y \geq 0 \end{cases} \quad (9.17)$$

It is clear that t_{pos} and t_{neg} identify portions of positive and negative trends respectively between the controller and process outputs in the time series data. Based on our conceptual understanding of the characteristics of sluggish control loops, idle index I_i is defined as:

$$I_i = \frac{t_{pos} - t_{neg}}{t_{pos} + t_{neg}} \quad (9.18)$$

The value of I_i lies between $[-1 \ 1]$. Extreme values of -1 and 1 result when t_{pos} and t_{neg} are 0 respectively. A positive value close to one implies that the controller is sluggish while a value close to -1 may be due to a well-tuned controller or oscillatory process output. Typically, control loops with idle index values between -0.4 to 0.4 indicate a well tuned controller.

The approach discussed above can be easily implemented both as online and off-line procedures. However, care should be taken when applying this method as: (i) it cannot distinguish between well-tuned controllers and the ones that lead to oscillations in the output; idle index in both cases will be close to -1 and, (iii) it can indicate that the performance is adequate for controllers that result in overshoot when subjected to load changes.

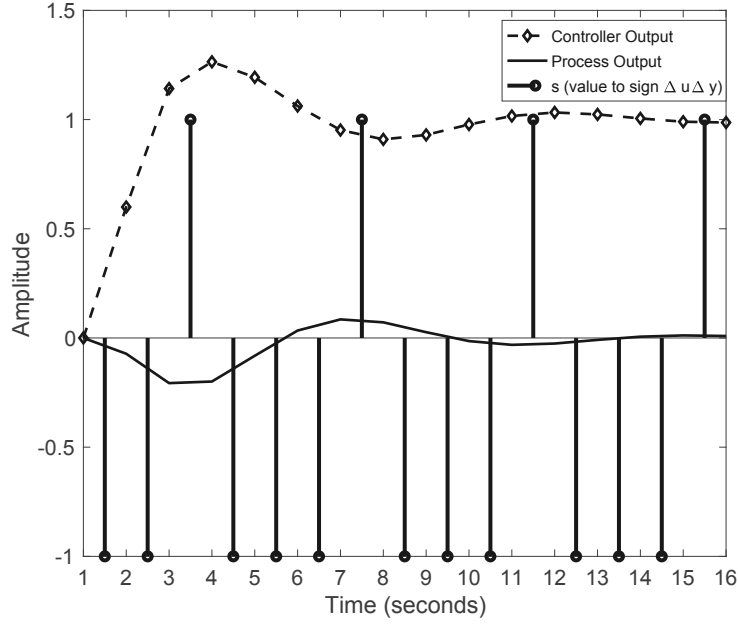


FIGURE 9.9: Response for PID controller setting $K_c = 2.5$, $\tau_I = 1$, and $\tau_D = 1$

Let us consider the previously discussed system $G(s) = \frac{1}{(s+1)^3}$ controlled with PID controller with settings $K_c = 2.5$, $\tau_I = 1$ and $\tau_D = 1$. A unit load disturbance in the process input is introduced and the responses obtained are shown in Figure 9.9.

Both Δu and Δy (change in controller and process outputs) are calculated and a sign value sgn is assigned as follows:

$$sgn = \begin{cases} 1 & \text{if } \Delta u \Delta y > 0 \\ -1 & \text{if } \Delta u \Delta y < 0 \\ 0 & \text{if } \Delta u \Delta y = 0 \end{cases} \quad (9.19)$$

The values of sgn for this example are shown in Figure 9.9. The value of the t_{pos} in this case is 4 while t_{neg} is 11. The value of the idle index is -0.47 indicating that the controller in this particular loop is well-tuned. Now, for the same process with different controller settings $K_c = 0.5$, $\tau_I = 20$ and $\tau_D = 2$ the process output, controller output and values of sgn are shown in Figure 9.10. The value of t_{pos} in this case is 44 while t_{neg} is 6. This is clearly

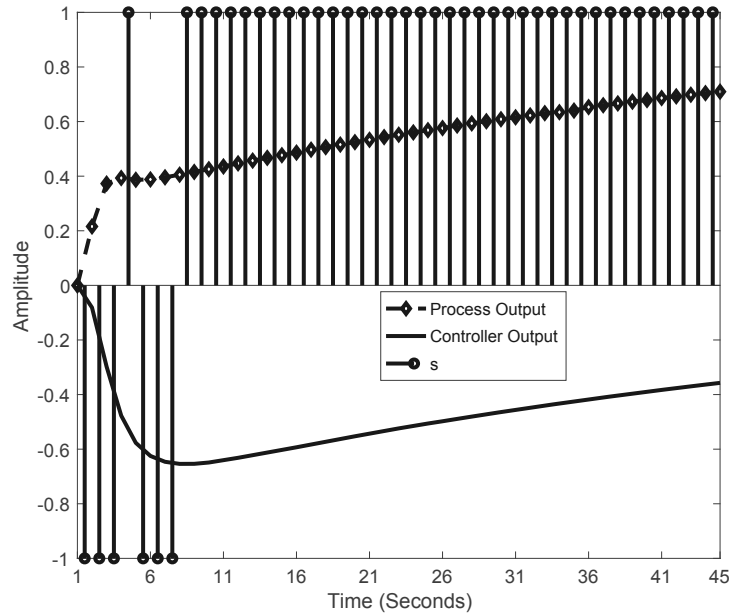


FIGURE 9.10: Response for PID controller settings $K_c = 0.5$, $\tau_I = 20$, and $\tau_D = 2$

a case where the controller is sluggish and an idle index of 0.76 also indicates the same.

9.4 Time Series Analysis: Preliminaries

Time series concepts are required to describe algorithms that can handle cases three and four in CPA. In this section, we will introduce the most relevant concepts that are required to understand the material that follows. Cases one and two addressed step changes in setpoint and disturbance variables respectively. These changes are deterministic. However, in control loop analysis, most of the disturbances are, typically, stochastic. As far as this book is concerned, we will simply understand stochastic behaviour as behaviour that is difficult to predict. This brings with it the associated difficulty of assessing the impact of variables that are unpredictable in control loop performance monitoring. Without being very mathematically detailed, we will explain these concepts to as much detail as required for one to understand the rest of the material.

From now on, we will think of data for the different variables such as y , e and so on as time series data (time stamped data as described before). Our interest lies in working with and developing models for these time series data. Consider a time series y of length N (that is, there are N time stamped data points). In all of the discussions from now on, it is assumed that we possess large amounts of data (N very large); with less data, the quantities that are computed become approximate. One of the fundamental questions that is of interest from a controller performance monitoring viewpoint is: how much variation is there in the values of the time series y . At the two ends of the spectrum are: y that is constant and y that is completely random. To identify if y is constant, we perform a simple calculation, which is the computation of the **mean** ($\bar{y}$) of the time series as shown below:

$$\bar{y} = \frac{\sum_{k=0}^{N-1} y(k)}{N} \quad (9.20)$$

If the time series is constant, then $y(k) - \bar{y} = 0 \forall k$. We will wait till the definition of other statistical properties before describing the other end of the spectrum, that is, a completely random or pure “white noise” signal (common nomenclature). If the time series is not constant, the amount of variability in the time series is computed through another quantity, called the variance. Variance quantifies the deviation of the signal from its mean value ($\bar{y}$) and is computed as shown below:

$$\sigma^2 = \frac{\sum_{k=0}^{N-1} (y(k) - \bar{y})(y(k) - \bar{y})}{N - 1} \quad (9.21)$$

σ is called the **standard deviation** and **variance** is the square of standard deviation. For now, let us understand mean and variance as some statistical properties of the time series and there are several such statistical properties that can be computed for a time series.

From a control system viewpoint, the reason we are interested in these properties is because the performance of the controller can be quantified using such properties. For example, if we are interested in the output reaching a setpoint, we compute the mean of the output time series. If the mean is close to the setpoint, then we know that our control objective has been reasonably met. Variance indicates how much the output is varying around this setpoint. A very large variance might be a cause for concern because this means that, while we have been able to move the output to the vicinity of the setpoint, high variance indicates that the output is not close to the setpoint at all times.

Another property of time series that we are interested in is **periodicity**. Time series data is said to exhibit periodicity of period T_P if $y(k + T_P) = y(k) \forall k$. Examples of periodic signals are sine and cosine functions. It is obvious that periodic signals are of relevance to control loop monitoring because

oscillations (discussed in the control context in several of the previous chapters) are comprised of periodic components.

Another property that is of interest can be understood as a slight variation of Equation (9.21). It is:

$$\sigma_{yy}(l) = \frac{1}{N-l} \sum_{k=0}^{N-1-l} ((y(k+l) - \bar{y})(y(k) - \bar{y})) \quad (9.22)$$

This quantity is called the auto-covariance function (ACVF) at lag l . The reason for the introduction of yy in the subscript is to generalize the equation so that it is applicable to two time series variables at the same time. If one were to use y and u in the equation with $\bar{y}$ and $\bar{u}$ as their respective means, we would have defined $\sigma_{uy}(l)$, which is the cross-variance between y and u at lag l . Further, it can be shown that the maximum value of $\sigma_{yy}(l)$ is for $l = 0$.

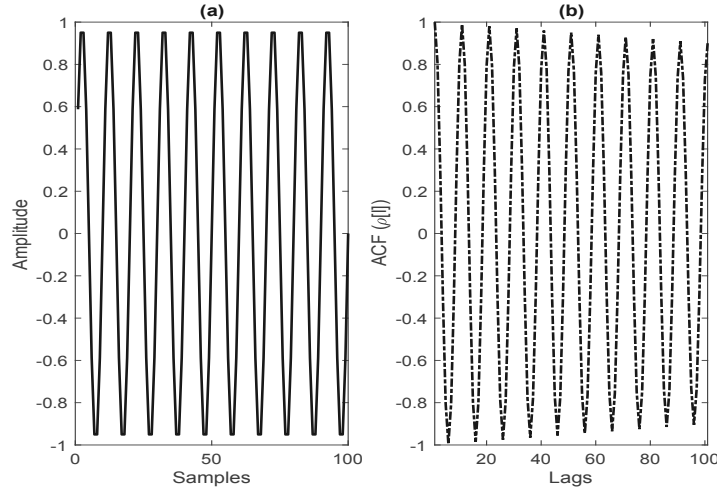
The three time series signals: constant, purely random and periodic can now be described in terms of $\sigma_{yy}(l)$. For constant signals $\sigma_{yy}(l) = 0 \forall l$, for periodic signals $\sigma_{yy}(l) = \sigma_{yy}(l + T_P)$. The observations regarding the constant and periodic signals are easy to verify from the definition shown in Equation (9.22). Interestingly, if the signal is periodic, then the auto-covariance function is also periodic with the same period. This property is very useful in controller performance monitoring. Finally, if y is a purely random or a “white noise” signal, $\sigma_{yy}(l) = 0 \forall l \neq 0$ and the only non-zero value is at $l = 0$, which will also be the maximum value as described before. This definition of white noise can be understood intuitively, if one views non-zero $\sigma_{yy}(l)$ as correlation between samples with a lag of l . If the time series is purely random, then one would expect no correlation between any of the samples; this is represented by the condition $\sigma_{yy}(l) = 0 \forall l \neq 0$.

One can also define a normalized version of Equation (9.22) as below:

$$\rho(l) = \frac{\sigma_{yy}(l)}{\sigma_{yy}(0)} \quad (9.23)$$

$\rho(l)$ is called the auto-correlation function (ACF) at lag l . It can be observed that $\sigma_{yy}(0)$ is the variance that was defined in Equation (9.21) and $\rho(0) = 1$.

Let us verify some of these theoretical discussions through simulations. Consider a periodic signal and a plot of the ACF (just a normalized version of ACVF) of the signal shown in Figure 9.11. One can observe that the peaks of these two signals match demonstrating that ACF has the same period as the original signal. We described that the ACF of white noise has a non-zero value only at $l = 0$. A white noise signal and the ACF of that signal are shown in the top panel of Figure 9.12. It can be seen that ACF is non-zero only at lag $l = 0$ and the ACF at other lags are very small. These should have ideally been zero but small errors are introduced due to the finite data length (N not large enough).

FIGURE 9.11: (a) Simulated data $x[k]$, and (b) ACF of $x[k]$ - periodic signal

In many cases, the original time series might be corrupted with purely random errors (or white noise); this happens due to random errors in measurements. In such cases, the properties of the original signal might become difficult to recover. For example, even if the original signal is periodic, it might be difficult to identify the period directly from the signal. We have discussed earlier that the ACF of the signal will also be periodic. The ACF of the signal might reveal the periodicity in the signal better than the signal itself. The bottom panel of Figure 9.12 demonstrates this where a sinusoidal signal corrupted with noise and the ACF of this signal are shown. The periodicity of the signal is clearly visible in the ACF plot. Thus, ACF can be used to obtain hidden oscillations (periodicity) in the noisy data and this idea is used in several algorithms for CPA.

There is one final concept that will be introduced before CPA algorithms are described. In all the equations that we showed before, it was assumed that the statistical properties of the signal remain the same (with time or samples). This assumption is referred to as the stationarity assumption. Time series signals that violate this assumption, where the statistical properties vary over time (or samples) are called non-stationary signals. Theoretical arguments that underlie time series analysis can be quite complicated and we restrict ourselves to this simple presentation that is adequate and of direct relevance to controller performance monitoring. Comprehensive material on time series analysis can be found in [51, 74, 93, 102].

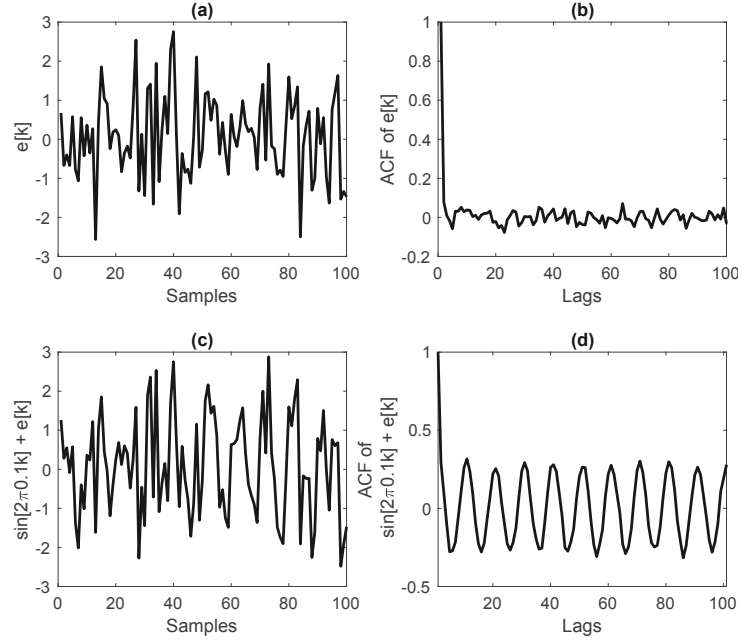


FIGURE 9.12: (a) Simulated white noise data $e[k]$ from Normal distribution using `randn` command in MATLAB[®], (b) ACF of $e[k]$, (c) Simulated data $\sin[2\pi f k] + e[k]$, and (d) ACF of $\sin[2\pi f k] + e[k]$

9.5 Identifying Increased Variability

The third characteristic (Figure 9.2) that we want to identify is increased variability. Variability is directly related to the variance that was described by Equation (9.21). However, since we are generally interested in the output following the setpoint, we modify the variance equation to the one shown in Equation (9.24):

$$\sigma_Y^2 = \frac{\sum_{i=0}^{N-1} (SP(i) - PV(i))^2}{N} \quad (9.24)$$

This quantity provides a measure of the variability of the process variable around setpoint. If control was perfect, variability will be zero. It is tempting to use variability directly as a measure to assess controller performance. This would assume that a controller can address all of the variability seen in the data. However, this would be incorrect and how much of this variability is

addressable by a controller is not an easy question to answer. In every process, there is a certain amount of irreducible variability and it would be unrealistic to expect controller redesign to address this variability. Understanding the reason for this irreducible variability (or variance) in control loops and as a consequence, developing algorithms that quantify this increased variability over what is normal is critical. This increased variability will determine the “upside potential” for control. Two approaches will be introduced in this chapter to quantify this opportunity for better control. The first approach is one where the minimum variance that a best performing controller can achieve is computed. Any variance over and above this, as a result, can be flagged as a control loop problem. The second approach is based on the idea of identifying repeating patterns, which implies that the signal is not truly random and hence some portion of the variance is reducible through better control.

9.5.1 Minimum Variance Index

Let us consider an FOPTD system with the following transfer function:

$$G_p(s) = \frac{3e^{-10s}}{10s + 1} \quad (9.25)$$

The PI controller transfer function is given by:

$$C(s) = 0.58 + \frac{0.01}{s} \quad (9.26)$$

The process output without any disturbance (case I) and the same process output corrupted with two different noises (non-step type disturbances), called case II and case III are shown in Figure 9.13. A white noise is added to the output in case II and a white noise is sent through a transfer function (also called filter) $G_d(s) = \frac{1}{10s+1}$ in case III. The variance of the signal after going through the filter is 0.006. Signals such as the one just described are termed as “colored noise” to denote that the signal is not purely random, but has been colored through a filter (equivalent to white light passing through a filter to produce colors). Let us keep in mind that this information is not known to the control engineer or the algorithm; only *PV* and *SP* data are available for analysis.

Given this data, the objective is to verify if the controller is tuned optimally to reject these two disturbances. To demonstrate the difficulty in answering this seemingly simple question, we first calculate the variances of the process outputs at steady-state after subtracting the setpoint values from the data. Variances of the data starting from 150 secs were calculated for cases II and III, and the variance values are 0.26 and 0.03, respectively. If one were to look at just the variance, then the conclusion that will be drawn is that there is significantly more “upside potential” for case II when compared to case III. To verify this, two “better” controllers are designed and the variances with the new controllers are computed.

The controller transfer function is modified to $C(s) = 0.1 + \frac{0.01}{s}$. The process variables corresponding to cases II and III with new controller settings are plotted in Figure 9.14. With this new data, the variance for case II is 0.21, while the variance for Case III is 0.007. Notice that in case III there is a variance reduction of 76%, while the variance reduction in case II is considerably less. To ensure that this is not a result of the new controller design, another controller transfer function $C(s) = 0.2 + \frac{0.02}{s}$ is evaluated. The variance for case II with this controller is 0.22, while that of case III is 0.008 (reduction of 73%). This example demonstrates that **the percentage improvement possible cannot be determined by purely looking at the variance data**. If this is the case, then, how does one identify the “upside potential” for each loop or how does one identify poorly performing loops, where intervention is likely to be beneficial?

One approach to answer the questions posed above is to resort to modeling the time series data. We simplify the presentation here greatly to just explain the basic ideas behind this approach. Detailed derivations and description of a solution to this problem can be found in [33]. Let us consider the time series data $\hat{y} = y - y_{sp}$. In many situations, time series data is modeled as pure white noise signal passing through an appropriate filter as shown in Figure 9.15. As

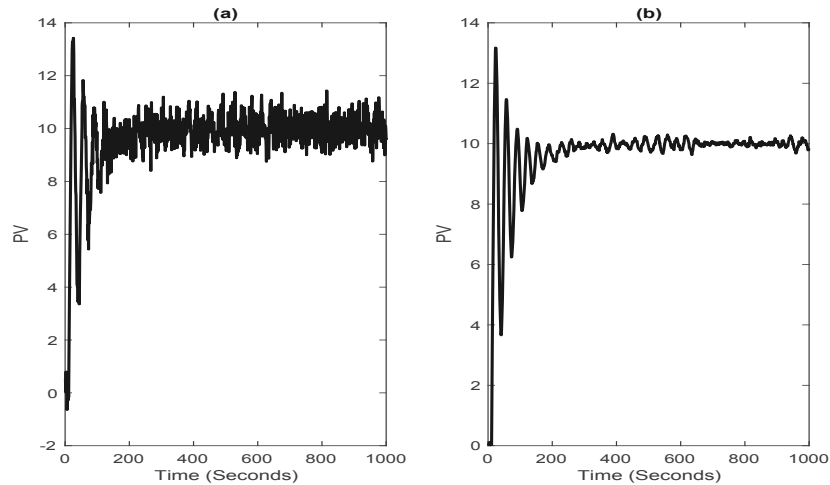


FIGURE 9.13: (a) Case II - Process output from closed-loop system with disturbance (not of step type), and (b) Case III - Process output from closed-loop system with disturbance (not of step type)

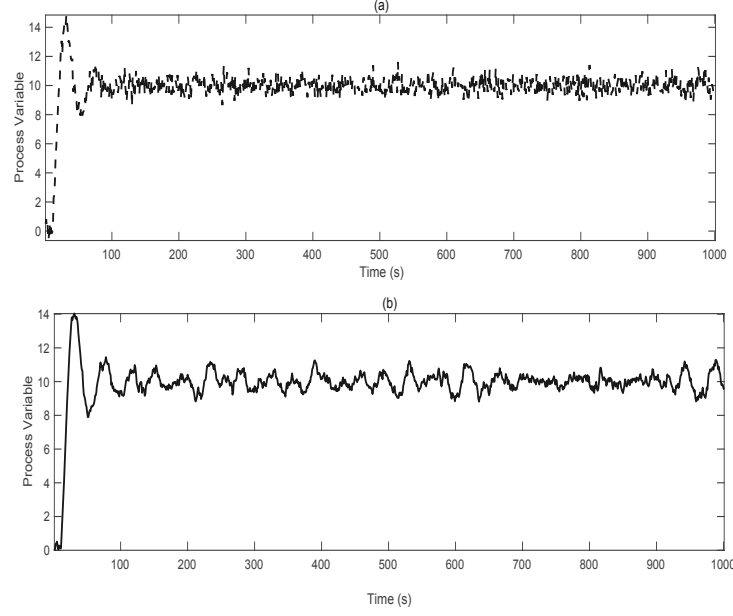


FIGURE 9.14: (a) Case II - Process output from closed loop system with new controller settings, and (b) Case III - Process output from closed-loop system with new controller setting.

a result, the effect of the disturbance (e) on the output $\hat{y}$ can be modeled through an FIR model (Equation (8.16)) as shown below:

$$\begin{aligned} y[k] - y_{sp}[k] &= \hat{y}[k] \\ &= e[k] + a_1 e[k-1] + \dots + a_{(\tau_d-1)} e[k - (\tau_d - 1)] \\ &\quad + a_{(\tau_d)} e[k - (\tau_d)] \dots + a_n e[k - n] \end{aligned} \quad (9.27)$$

We are interested in the variance of $\hat{y}$, which can be computed through:

$$\sigma_{\hat{y}}^2 = \left(1 + a_1^2 + a_2^2 + \dots + a_{(\tau_d-1)}^2 + \dots + a_n^2 \right) \sigma_e^2 \quad (9.28)$$

In the equation, σ_e^2 is the variance of the white noise variable $e(t)$; errors at different time instants are assumed to be independent and identically distributed. Equation (9.28) can be split into two components as below:

$$\sigma_{\hat{y}}^2 = \left(1 + a_1^2 + a_2^2 + \dots + a_{(\tau_d-1)}^2 \right) \sigma_e^2 + (a_{\tau_d}^2 + \dots + a_n^2) \sigma_e^2 \quad (9.29)$$



FIGURE 9.15: Time series modeling for MVI computation

At this point, knowledge of time-delay τ_d allows us to compute an irreducible variance in the output. If there is a delay of τ_d , control actions implemented only up to $k - \tau_d$ time instants can have an impact on y . Observing Equation (9.27), it is clear that terms until $a_{(\tau_d-1)}e[k - (\tau_d - 1)]$ cannot be compensated for by any control action. The best control action can only remove the terms including and after the $a_{(\tau_d)}e[k - (\tau_d)]$ term. As a result, the irreducible variance in $\hat{y}$ will be the first term of the two terms in Equation (9.29). Based on this, a ratio is computed as a performance index as shown below (σ_e^2 will cancel out):

$$MVI = \frac{\left(1 + a_1^2 + a_2^2 + \dots + a_{(\tau_d-1)}^2\right) \sigma_e^2}{\left(1 + a_1^2 + a_2^2 + \dots + a_{(\tau_d-1)}^2 + a_{\tau_d}^2 + \dots + a_n^2\right) \sigma_e^2} \quad (9.30)$$

This index is called Minimum Variance Index (MVI). It can be seen that the maximum value of this index is one. At this value, the interpretation is that all the variance that is seen in the output is irreducible. Consequently, the performance of the controller cannot be improved by further tuning. However, MVI close to one might be accompanied by large control actions resulting in wearing of control valves. Therefore, controllers are tuned such that MVI is between 0.8 – 1. Any value of MVI less than 0.5 indicates significant degradation in control loop performance and therefore steps to improve performance should be undertaken.

In Equation (9.27), both the parameters of the model and the inputs ($e[k - 1] \dots e[k - n]$) are not known, unlike the model in the MPC chapter where the input is assumed to be known. At this point, we will only say that there are algorithms that can estimate coefficients of models of the form shown in Equation (9.27). A simple code in MATLAB[®] that can perform this model building is provided after a summary of the MVI computation procedure.

- Obtain process data $y[k]$ with variations around the setpoint. Compute $\hat{y}[k] = y[k] - y_{sp}$
- Model $\hat{y}[k]$ as a function of white noise as in Equation (9.27). Choose model order (number of coefficients) to be at least $\tau_d + 10$. Modeling is

typically performed using ‘armax’ command in MATLAB®. Model coefficients a_1 to a_n are then obtained by a simple long division procedure

- Compute MVI using Equation (9.30). This requires the knowledge of process delay

The MATLAB code for computation of MVI is given as follows:

```

1 function minvar2(yprime,d,Ts)
2 y = iddata(yprime,[],Ts);
3 % d = time delay
4 %Ts = Sampling time
5 th = armax(y,[d+10 d+10]); % Arma model of %output
6
7 C=th.C;
8 A=th.A;
9 f=ldiv(C,A,(d-1)); % long division to get %coefficients
10 yhat=predict(th,y,1); % prediction of output using %model
11 ahat=y.y-yhat.y; % compute residuals (noise %variance)
12 for k=0:d-2
13     y2=y.y(k+1:end);
14     a2=ahat(1:end-k);
15 end
16 mvar=0;
17 for i=1:d-1
18     mvar=mvar+(f(i)^2)*var(ahat);
19 end
20 mvar
21 %computation of total variance
22 yvar=var(y.y)
23
24 MVI=mvar/yvar

```

Let us revisit case III of the example described at the beginning of this section to verify the theoretical MVI approach. The process output for this case is shown in the bottom panel of Figure 9.13, and the controller transfer function is given by $C(s) = 0.34 + \frac{0.04}{s}$. armax command in MATLAB® (code above) with model order 20 (10 coefficients more than process delay of 10) is used to model this data after removal of set-point. An MVI value of 0.3 is computed for this case, which indicates that there is considerable scope for improvement through controller redesign. This hypothesis can be verified through the new controller (described in that example) $C(s) = 0.1 + \frac{0.01}{s}$. The MVI value increases to 0.9 indicating that this new controller is better and with this controller in place, there might not be further scope for improvement. In contrast, MVI for case II with the original controller is 0.8, implying that there might not be significant scope for improvement.

9.5.2 Scaling Exponent Based Measure

In the MVI approach, accurate knowledge of process delay is a critical requirement. Let us take reconsider case III (discussed before) with $C(s) = 0.1 + \frac{0.01}{s}$, and compute MVI assuming that the delay is 2 instead of 10. The new MVI value is 0.24, which will indicate that the controller is not optimal. However, we know this not to be true. A delay information free performance measure might be an attractive proposition for controller performance assessment.

A method based on scaling exponent, which does not require delay information is described next. The intuitive idea behind this approach is the following. If there are components in the time series that are **amenable** to modeling, then we assume that they are also amenable to control. One way to think about modelability is to look for repeating patterns in the signal. If repeating patterns exist, then they will lead to some form of self-similarity. Self-similarity as a concept has been used to describe objects. For objects, self-similarity implies that an object is similar to itself at different sub-unit levels. When one extends this idea to time series data, then one might look for self-similarity in statistical properties.

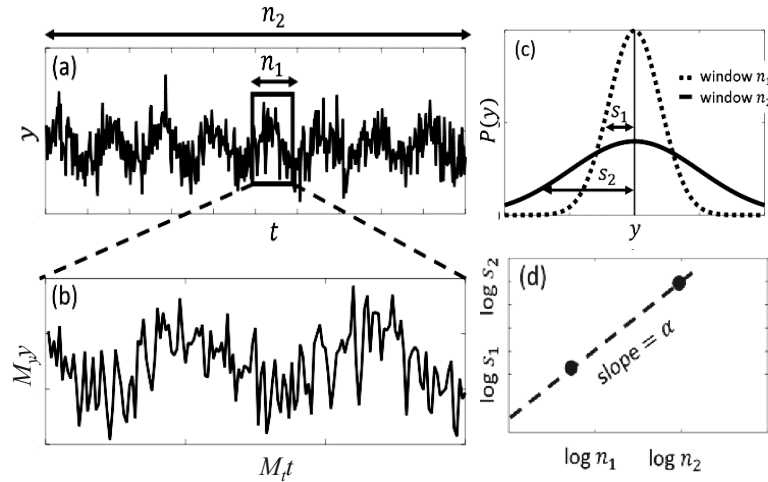


FIGURE 9.16: Self-similarity of a time series (adapted from [99])

For a time series, shown in Figure 9.16, self-similarity can be assessed by sub-selecting a portion of the data and verifying if the statistical properties of this sub-selected sample is the same as the whole data. However, since the scales are different, similarity of statistical properties are assessed after

scaling ‘t’ and ‘y’ axes. With appropriate scaling, a time series is said to be “statistically similar” to its scaled version $a^\alpha y\left(\frac{t}{a}\right)$ if [99]:

$$y(t) \cong a^\alpha y\left(\frac{t}{a}\right) \quad (9.31)$$

Both ‘t’ and ‘y’ are scaled for the sub-selected portion of the data and the equality is verified for all statistical properties (such as mean, variance etc.,) of the original time series and its scaled version. For our purposes, it is enough to ensure that the mean and variance of the original time series $y(t)$ is identical to that of its scaled version $a^\alpha y\left(\frac{t}{a}\right)$. If we denote ‘t’ scaling by M_t and ‘y’ scaling by M_y , then we have:

$$\alpha = \frac{\ln(M_y)}{\ln(M_t)} \quad (9.32)$$

A simple scaling for ‘t’ is $M_t = \frac{n_2}{n_1}$, where n_2 is the duration (window length) of the original time series and n_1 indicates the duration of the subset of the time series (see Figure 9.16). It is easy to verify that using a scale factor of ‘y’, which is $M_y = \frac{s_2}{s_1}$, where s_2 and s_1 denote the standard deviations of the original and sub-unit time series respectively, will ensure that the variance of the original and the scaled signals are the same. Using these definitions for M_t and M_y , one can compute α , given by:

$$\alpha = \frac{\ln s_2 - \ln s_1}{\ln n_2 - \ln n_1} \quad (9.33)$$

In general, one would not directly compute α based on one sub-segment of data but compute it using multiple sub-segments of different lengths as a slope of a log-log plot. The α value is also known as the scaling exponent of the time series. Given a discrete time series, there are several ways to find the scaling exponent and one popular algorithm is known as “Detrended Fluctuation Analysis (DFA)” [45, 65]. This algorithm is applied on an integrated signal (making the signal non-stationary) as described below (adapted from [83]):

1. Transform the given time series $y[k]$ to $y'[k] = \sum_{k'=1}^k (y[k'] - \bar{y})$. Here $\bar{y}$ is the mean of $y[k]$. Both $y[k]$ and $y'[k]$ will be of length N
2. The integrated time series $y'[k]$ is segmented into W windows of length L ($WL = N$). In each window, a least squares fit is performed using the objective function $\min_{a_j, b_j} \epsilon_j^2 = \sum_{i=1}^L (y'[i] - a_j i - b_j)^2$
3. The root mean square (rms) fluctuation is calculated for all W windows using the formula:

$$F(L) = \sqrt{\frac{1}{WL} \sum_{j=1}^W \epsilon_j^2}$$

4. Repeat steps 2-3, varying the window length (L) between a minimum length of 10 samples and a maximum length of $N/4$ with N being the number of samples
5. A log-log plot of the rms fluctuations against the corresponding window lengths should yield a straight line. The slope of this graph is the generalized Hurst exponent α

9.5.2.1 Implications of the Scaling Exponent to Control Loop Performance Assessment

The value of α for a time series with approximately constant mean and variance ranges between 0 and 1 with $\alpha > 0.5$, indicating the presence of long-term correlations in the data and $\alpha < 0.5$ indicating the presence of short term/anti-correlation in the data. For any time series with no correlation in the data, the value of $\alpha = 0.5$. Scaling exponent value $\alpha > 1$ indicate that the time series has a time varying mean and variance. Deviation of α from 0.5 in either direction indicates possible presence of modelable components. A scaling exponent value of 0.5 is not exactly achievable by any controller. However, significant deviations of α from 0.5 indicate “upside potential” for controller redesign.

A generalized scaling exponent based performance measure which has a range of 0-1 similar to the MVI is defined as [83]:

$$\eta = \begin{cases} \frac{\alpha}{0.5} & \text{if } \alpha \leq 0.5 \\ 1.5 - \alpha & \text{if } \alpha > 0.5 \end{cases} \quad (9.34)$$

The value of η lies in the range of 0–1 with a value close to one indicating good/satisfactory control loop performance, while significant deviations from one indicates potential for improvement.

Revisiting case III with $C(s) = 0.2 + \frac{0.02}{s}$, the scaling exponent value using DFA is 0.66 and an η value of 0.84 indicating satisfactory performance of the control loop. For case III with $C(s) = 0.58 + \frac{0.01}{s}$, the value of η is 0.58 indicating poor performance of the control loop in rejecting non-step type disturbances. In contrast, for case II with both the controller settings, the value of η lies between 0.8 - 1 indicating that there is very little scope for improvement.

9.6 Detecting Oscillations

The last feature (and perhaps the most important one) that we are interested in control loop performance monitoring are oscillations in data. A number of methods have been proposed in the literature for detecting oscillations [23, 31, 40, 41, 82, 94, 95]. These methods can be broadly classified as described below.

- Detection using time-domain properties like Integral Absolute Error (IAE)
- Detection based on auto-correlation function (ACF)
- Detection based on spectral properties of the system
- Detection using advanced techniques like wavelet transforms

The order of complexity of the solution increases as we traverse down the list. Each solution has its own merits and demerits. In the following section we will describe a method called decay-ratio approach for oscillation detection.

9.6.1 Decay-ratio Approach for Oscillation Detection

In the chapter on process identification, a quantity called decay-ratio was defined for oscillating signals. The decay-ratio quantifies the rate at which oscillations vanish. *If the controller is well tuned, then oscillations should decay rapidly.* Rapidly dwindling oscillation will have a low decay value. Since one is interested in quantifying the performance of control loops from an oscillations viewpoint, a slightly modified definition of the decay-ratio (measure called R) is suggested to detect oscillations [55]:

$$R = \frac{a}{b} \quad (9.35)$$

where a is the distance between first maxima and the line connecting first two minima of the output, b is the distance between the first minima and the line joining the output at the first sample and the first maxima. This is shown in Figure 9.17. The value of R is computed from the ACF function of the output instead of the actual output. We have already discussed that ACF is of the same periodicity as the original signal and further, periodicity might be more pronounced in the ACF signal than the original signal (Section 9.4)

A value of $R = 0.5$ is proposed as a threshold for detection of oscillations in [55]. The value of $R \geq 0.5$ can be interpreted as the signal exhibiting excess degree of oscillations. The threshold value can be chosen depending on the application. This procedure can be implemented using the algorithm given below (assuming that certain minimum data length criteria are met).

1. Compute the ACF of the given data $y[k]$

TABLE 9.1: Data for simulating oscillations

Test Case	τ	ω	σ_e^2	R
Oscillatory case	5.6	6	0.05	0.64
Non oscillatory case	1	6	0.05	0.39

2. Check for presence of at least two minima and one maxima in the ACF. If there are less than two minima in the ACF, set $R = 0$ and go to step 5
3. Calculate a (distance between first maxima and the line connecting first two minima of the ACF) and b (distance between the first minima and the line joining the output at the first sample and the first maxima in ACF) as shown in Figure 9.17
4. Compute the decay-ratio $R = a/b$
5. If $R < 0.5$, $x[k]$ does not have significant oscillations. Otherwise, it is oscillatory

To numerically verify the R measure, let us consider a signal generated using the following equation:

$$y[k] = e^{-\frac{k}{\tau}} \sin[2\omega k] + e[k] \quad (9.36)$$

where $e[k] \sim \mathcal{N}(0, \sigma_e^2)$, where $\mathcal{N}$ is a normal distribution with zero mean and variance, σ_e^2 . Appropriate values are assumed to simulate oscillatory and non-oscillatory case as shown in Table 9.1.

9.7 Summary

In this chapter, ideas related to monitoring control loop performance once it is in operation were discussed. Typical problems seen in control loops were categorized and techniques for monitoring of control loop data to identify these problems were described. This is an evolving field, very rich in contributions in the last couple of decades. There are several advanced ideas in SISO control loop monitoring, MIMO loop monitoring, and MPC performance monitoring that are important but not covered here. For performance assessment of multivariate systems and other concepts, interested readers are referred to [3, 5, 16, 17, 18, 38, 42, 77, 80]. There are also several other approaches for oscillation detection and characterization that have been explored in the literature [15, 30, 50, 81, 87, 95].